

Differential game model of carbon emission reduction decisions with two types of government contracts: Green funding and green technology

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ABSTRACT

In this study, two differential pollution games are constructed according to the impacts of green technology and green funding on promoting carbon emission reduction. In the game, the government provides reasonable green funds or green technologies to encourage enterprises to reduce emissions. Enterprises choose the optimal production plan that maximizes their own interests according to the incentive mechanism of the government. We compare green funding and green technology in terms of total emissions, reduction efficiency and corporate profits. The results demonstrate that: (1) government contracts are not always beneficial for enterprises reducing emissions; (2) when the coverage of green funding is greater than a certain threshold, the emission reduction efficiency brought by green funds is better than that of green technologies; (3) when the funding intensity exceeds a certain value, there exists an optimal coverage of green funding to ensure that corporate profit targets and emission reduction targets are met simultaneously. The results provide a theoretical basis for the government to formulate an emission reduction incentive mechanism.

1. Introduction

Climate change is still intensifying. According to the interim report “State of the Global Climate 2021 – Extreme Events and Major Impacts” released by the World Meteorological Organization (WMO) on the opening day of COP26, global temperatures have been the highest on record over the past seven years. More and more scientific evidence shows that climate change, characterized by global warming, has become an indisputable fact. (Meinshausen et al. (2009)). Greenhouse gas emissions from human activities, especially fossil fuel consumption, are believed to be the main cause of climate change (Schelling (1984)).

However, climate is a public good and, as a result, a single carbon-dioxide emitters, such as a country or a region or a enterprise, has no incentive to reduce greenhouse gas emissions if other carbon-dioxide emitters do not take mitigation actions (Wei et al. (2015)). In addition, greenhouse gases are geographically diffused, and unilateral responses from a single region are often ineffective. Regional cooperation in pollution control is an effective way to reduce pollutants (Huang et al. (2016)). In many papers, the government is considered to be the decision maker that can determine regional output and emission reduction costs to achieve lower emissions by setting positive emission reduction incentives (Hong et al. (2017)). Industrial enterprises are the main

producers of greenhouse gas emissions, so it is necessary to make different production plans when facing different incentive policies of the government (Wei and Wang (2021); Chen and Zhu (2022); Chen et al. (2022)).

The positive incentive contract implemented by the government mainly includes green funding and green technology to make up for the high cost of green innovation (Yi et al. (2021), Chen et al. (2021) and Chen et al. (2022)). The main purpose of this paper is to compare the advantages and disadvantages of green funding and green technology in promoting emission reduction and try to find the applicable scope of them.

In order to achieve the above purposes, this paper considers differential transboundary pollution game model composed of a government and N homogeneous enterprises. In the game, the government provides positive green incentive contracts to encourage green innovation and achieve emission reduction targets. Enterprises choose different production plans according to different incentive contracts. The decision-making behaviors of both of them jointly affect the amount of greenhouse gas accumulation. Two differential transboundary pollution game are built according to the impacts of green technology and green funding on promoting emission reduction. Then, green funding and green technology are compared in terms of optimal emissions, pollution control,

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and corporate returns etc. The applicable scope and mutual transformation conditions of green funding and green technology are also discussed.

The main contributions of this paper are as follows: Firstly, both green funding and green technology can effectively motivate enterprises to reduce emissions. Specifically, in the case where green funds and green technologies are equivalent, the emission reduction efficiency brought by green funds is better than that of green technologies when the coverage of green funding is greater than a certain threshold; the total emission reduction brought by green funding is greater than that of green technology when the funding intensity exceeds a certain value. Otherwise, green technology is superior to green funding.

Secondly, we find the key condition for enterprises and the government to choose green funding in our model: there exists a optimal coverage of green funding to ensure that targets of corporate profit and emission reduction could be met simultaneously when the funding intensity exceeds a certain value. This condition can avoid the failure of green technology transfer due to intellectual property rights or other reasons, which will affect the development of emission reduction.

The rest of this paper is organized as follows: Section 2 reviews the related literature. Section 3 gives the differential transboundary pollution game model, and Section 4 describes the feedback Nash equilibria of the two games. A comparison of the two networks from the perspectives of pollution control and mitigation costs is presented in Section 5. Section 6 provides a numerical study, and Section 7 concludes the paper.

2. Literature review

Differential game are an effective tool to deal with the pollution problem and are favored by many scholars because they can provide a good reference for the strategic behavior and dynamic pollution control of game participants in continuous system (Wang et al. (2022), Bertinelli et al. (2014), de Frutos and Martín-Herrán (2019) and Zhou and Ye (2018)). For example, Van Der Ploeg and de Zeeuw (1992), Long (1992) firstly introduced the differential game model into the transboundary pollution problem and obtained the cooperative optimal solution. Ploeg and Zeeuw (1992) provides a feedback equilibrium for the transnational pollution problem in the non-cooperative case, and a more general nonlinear Nash equilibrium solution is given in Dockner and Long (1993). Furthermore, Rubio and Casino (2002) compares the feedback solutions in cooperative and non-cooperative cases.

Adding some social and economic factors on the differential game model can deal with more realistic emission problems. For example, adding a stochastic process to the game model can handle the impact of uncertainty and contingencies on emissions reductions (Yeung and Petrosyan (2008), Jiang et al. (2019) and Liu et al. (2021)); the effect of learning by doing can effectively reduce the cost of emission reduction (Chang et al. (2018), Li and Pan); the introduction of a competition mechanism can better encourage enterprises to develop and use new technologies (El Ouardighi et al. (2021); Deng et al. (2021)); the introduction of an incentive mechanism can further improve the enthusiasm for low-carbon technology innovation in enterprises (Yang and Chen (2018), Li and Chen (2021) and Tuofu et al. (2022)).

From the perspective of governance mechanisms, the most common governance method is administrative means, that is, the government as a game party directly participating in environmental governance decisions (Wei and Wang (2021), Zaman and Zaccour (2021) and Yang et al. (2021)). For example, Benchekroun and Long (1998) analyzed the production strategy of oligarch enterprises facing the implementation of carbon tax by government under the dynamic Gounot game. Kennedy (1994) studied the effect of a carbon tax strategy and a carbon emission trading system on enterprise emission reduction. The authors found that firms were more likely to use new technologies if pollution damage was linear.

Although the above studies explored the optimal strategies for

transboundary pollution control from different perspectives, there are still some deficiencies.

Firstly, when dealing with the emission reduction problem, the above literature all assume that the game player has a strong willingness to reduce emissions and will participate in emission reduction spontaneously. However, this assumption is too ideal and does not match reality.

Second, researchers on emission reduction incentive mechanisms only consider green technology (Li et al. (2022); Wang et al. (2021)) or green funding (Ling et al. (2022); Cui et al. (2020)) or focus on the comparison between financial subsidies and carbon tax (Fikru (2022)), but they seldom pay attention to the comparison between green funding and green technology. By introducing the regulation factors of government subsidies into the production planning of enterprises, the optimal output and emission reduction strategies of enterprises under different types of subsidies are discussed, and the characteristics of green technology and green funding are compared in depth. Therefore, this paper can be regarded as an extension of the above work to some extent.

At last, lack of effectiveness analysis and comparison of incentive mechanisms. Which positive incentive mechanism, such as green finance or green technology, is more conducive to emission reduction? How to define the respective scope of application? None of these questions have been well answered.

3. Problem description

3.1. Green projects

In order to achieve the emission reduction target effectively, the Kyoto Protocol stipulates that relevant countries and regions can choose three flexible implementation mechanisms to accomplish the emission reduction task.

1. Clean development mechanism. This mechanism allows industrialized countries and regions to implement feasible green emission reduction projects in developing countries by providing financial and technical support. Its original intention is to achieve global emission reduction through cooperation between developed and developing countries on the one hand and to help developing countries to optimize and upgrade their industrial and energy structures on the other hand. CDM projects can bring benefits to investors and developing countries. However, the benefits brought by CDM projects to developed countries and developing countries are not equal, and there are many risk factors in CDM projects, resulting in relatively high transaction costs.
2. Joint performance mechanism. Many countries within developed countries can be regarded as a group alliance. According to the actual situation, some countries will reduce emissions, while some countries will increase emissions, so as to complete the total emission reduction task. Joint implementation is limited to project cooperation between developed countries. And both contracting parties have stipulated an emission reduction quota and realize the transfer of quotas through cooperation.
3. Carbon emissions trading scheme. If a certain country or region struggles to complete a given emission reduction task, it can purchase a certain amount of emission reduction credits from a country or region that has overfulfilled the emission reduction task through the trading market, then mortgage its own credit, in order to achieve the goal of reducing greenhouse gas emissions. There is a risk of price failure in the carbon trading mechanism, and emergencies, such as outbreaks of COVID-19 or energy shortages caused by local conflicts, will have a huge impact on carbon prices.

As a developing country, China has not yet fully established an independent carbon trading market, and most of the projects to reduce greenhouse gas emissions can only be done through the Clean

Development Mechanism. This paper takes local governments and enterprises as research objects to simulate and study the subsidy issue in CDM projects.

3.2. Game model

This paper will study two different types of green-incentive-contracts design problems without considering political factors and analyze the optimal production strategy that enterprises can achieve under the two dynamic green contracts provided by the government. The government, in turn, designs appropriate green contracts to strike a balance between local economic development and pollution control. As for enterprises, they have two strategies to choose: one is to be green enterprise with receiving contracts; one is to be a generic enterprise without receiving contracts. Compared with general enterprises, green enterprises have higher production costs and lower carbon emissions.

Consider that N homogeneous enterprises in a region produce the same product. It is assumed that the output produced by enterprise i is q and that the unit production cost is c , then the local total output Q satisfies $Q = \sum q = Nq$. The price of goods on the market is inversely proportional to the total output, so it can be written as $p = p_0 - aQ$, where $p_0 > c$.

Enterprises will produce a certain amount of carbon emissions in the production process, and the emissions will accumulate over time due to the existence of the "air basin" (Liu et al. (2022); Qin et al. (2015)), and the accumulation of air pollutants might be influenced by natural degradation (Liu et al. (2022)). Denoting the emission as e and the cumulative amount as $x(t)$, the pollution evolution equation can be expressed as:

$$\frac{dx(t)}{dt} = Ne - \delta x, \quad (1)$$

where $\delta \in [0, 1]$ measures the CO_2 absorption rate of nature.

Since carbon emissions are associated with production, we assume that $e = q$ if an enterprise is a generic enterprise; $e = \gamma q$ with $\gamma \in (0, 1)$ if an enterprise is a green enterprise.

The government provides two different types of green contracts, green technology or green funding, denoted as $T_T(t)$ and $T_C(t)$ as a continuous function over time. Furthermore, it should be noted that a technology can be provided to many enterprises at once, whereas a equivalent amount of capital can only be given to one enterprise.

4. The optimal solution of the game model

4.1. The case of green technology

Based on the discussion above, all enterprises in this case could receive green technology from the government. That is, in this scenario, all enterprises are green enterprises. These enterprises have higher production costs due to lower emission efficiency.

The cost of emission reduction, which refers to the cost of learning, using and maintaining the new technology, could be written as aq .

To sum up, the income of enterprises consists of four parts: commodity income, commodity production cost, emission reduction cost and green technology from the government.

Therefore, the net profit π of enterprise i could be expressed as

$$\begin{aligned} \pi_i &= pq - cq - aq + T_T q = \\ &= (p_0 - aQ)q + (T_T - c - a)q = \\ &= (p_0 - aNq)q + (T_T - c - a)q, \end{aligned} \quad (2)$$

That is, the optimization goal of the enterprise is

$$\max_q \{ (p_0 - aNq)q + (T_T - c - a)q \}. \quad (3)$$

The optimal output of enterprise i can be calculated as the first-order condition of equation (3), as follows

$$\frac{\partial \pi_i}{\partial q} = (p_0 + T_T - c - a) - 2aNq, \quad (4)$$

Which means the best output of enterprise i is

$$q^* = \frac{p_0 + T_T - c - a}{2Na}. \quad (5)$$

It can be seen that, when N becomes larger, the optimal output of enterprise i will decrease. In addition, the output of enterprises is affected by the subsidy rate and emission reduction cost coefficient: obviously, a higher subsidy rate leads to higher output, while a higher cost of reducing emissions leads to lower output.

The government's aim is to strike a balance between curbing carbon emissions and local economic benefits. For the sake of discussion, we assume that the damage caused by the emission accumulation to the region is a linear function of x , denoted as βx .

Thus, the government's revenue π_g function can be expressed as

$$\pi_g = Q^* - T_T q^* - \beta x,$$

where Q^* is the total output, and $T_T q^*$ is the total amount of government subsidy. $Q^* = Nq^*$ is the sum of enterprises' output in equilibrium state.

That is, the optimization goal of government is

$$\begin{aligned} \max_{T_T} \quad & \int_0^t [Q^* - T_T q^* - \beta x] dt \\ \text{s.t.} \quad & \\ \dot{x} \quad &= N\gamma q^* - \delta x, \end{aligned} \quad (6)$$

To solve optimal problem (6), the Hamilton–Jacobi–Bellman (HJB) equation V_T could be written as

$$rV(x) = \max_{T_T} \left\{ [Q^* - T_T q^* - \beta x] + \frac{\partial V}{\partial x} [N\gamma q^* - \delta x] \right\}. \quad (7)$$

Combining equation (5), the optimal subsidy strategy of government could be calculated as the first-order condition of equation (7):

$$\frac{\partial V(x)}{\partial T_T} = \frac{1}{2a} - \frac{\psi + 2T_T}{2Na} + \frac{\lambda \gamma}{2a} = 0,$$

which means

$$\lambda = \frac{2a}{\gamma} \left(\frac{\psi + 2T_T}{2Na} - \frac{1}{2a} \right), \quad (8)$$

where $\psi = p_0 - c - a$ and $\lambda = \frac{\partial V(x)}{\partial x}$ is the shadow price that represents the cumulative effect of one additional unit of emissions on the total benefit. In this study, it is the marginal effect of pollution x . Here, $\lambda < 0$ represents the emission reduction that is beneficial to enterprise profits and vice versa.

By using Pontryagin's maximum principle, the shadow price λ satisfies

$$\dot{\lambda} = (r - \delta)\lambda - 2\beta.$$

Take the partial derivative of x on both sides of formula (7), and combine it with Formula (8) to obtain

$$\left[\frac{T_T \psi}{2Na} - \frac{2\delta x}{N\gamma} \right] T'_T - \frac{(\delta + r)(2T_T + \psi)}{N\gamma} + \frac{r + \delta}{\gamma} - \beta = 0, \quad (9)$$

where $T'_T = \frac{\partial T_T}{\partial x}$.

This paper considers the case where subsidies are a linear function of cumulative emissions, i.e.,

$$T_T = k_T x + b_T. \quad (10)$$

Substituting formula (10) into equation (9), we have

$$\begin{cases} k_T = \frac{2a(r+2\delta)}{\gamma}, \\ b_T = \frac{-(r+3\delta)\psi - (r+\delta-\beta\gamma)N}{2\delta}, \end{cases} \quad (11)$$

Green technology in the form of equation (10) is known as a feedback Nash equilibrium solution. It not only changes with time but also is related to emission accumulation x .

In addition, according to condition (11), the green technology rate k is negatively correlated with the enterprise emission factor γ . In other words, the more efficient the reduction, the greater the amount of green technology needed. Furthermore, k is also related to a , that is, the demand for green technology for enterprise emission reduction will become larger if the commodity-demand sensitivity to price is high.

In addition, the emission reduction cost coefficient α , pollution cost coefficient β and the number of enterprises participating in emission reduction N will also affect the optimal green technology. However, when the cumulative emission amount of x is relatively large in the initial stage of emission reduction, the influence of these three factors on optimal technology transfer is far less than that of emission factor γ and price sensitivity a .

At last, in the case of green technology, substituting equation (5) into equation (3), we obtain the net profit of the enterprise selling products

$$\pi_i = \frac{(p_0 + T_T - c - \alpha)^2}{4Na}.$$

The following proposition gives the optimal plan of enterprises.

Proposition 1. In the green technology case, the optimal output q_T^* of each enterprise is

$$q_T^* = \frac{p_0 + T_T - C - \alpha^T}{2Na}.$$

Based on q_T^* , the net profit of each enterprise is

$$\pi_T^* = \frac{(p_0 + T_T - c - \alpha)^2}{4Na}.$$

The emission of each enterprise E_T^* is

$$E_T^* = \gamma * q_T^* = \frac{p_0 + T_T - C - \alpha^T}{2Na} \gamma.$$

The total emission TE_T^* in the game is

$$TE_T^* = NE_T^* = \frac{p_0 + T_T - C - \alpha^T}{2a} \gamma.$$

λ is the shadow price of the emission accumulation of enterprises, which means the loss of enterprises caused by the cumulative increase of unit emission. Therefore, the shadow price λ should be less than 0, since an increase of one unit of emissions will lead to a reduction in the total benefit.

$$\lambda = \frac{2a}{\gamma} \left(\frac{\psi + 2T_T}{2Na} - \frac{1}{2a} \right) < 0,$$

which means

$$T_T < \frac{N - \psi}{2}. \quad (12)$$

Equation (12) suggests that the relationship between T_T and the development of emission reduction is not a simple linear relationship.

- (1) When $0 < T_T < \frac{N - \psi}{2}$, $\lambda < 0$, which means increasing green technology appropriately could rapidly develop emission reductions.
- (2) When $T_T > \frac{N - \psi}{2}$, however, $\lambda > 0$, which means increasing green technology will be unfavourable to the development of emission reduction. Given increasing T_T , the percentage of green assistance within the enterprises' net profits will be greater. When it reaches

a certain level, an enterprise will cease to focus on its sales volume. This means that a lot of goods are produced but not put to market, resulting in a waste of resources.

As for the emission accumulation amount x , we expect it to show a declining trend under the technology transfer in the form of (10), i.e.

$$\dot{x} = N\gamma q^* - \delta x < 0,$$

which implies

$$N > N^* = \frac{(r + \delta)(4a\delta x - \psi)}{r + \delta - \beta\gamma},$$

which means that, in the case of green technology, the carbon-dioxide accumulation can be effectively controlled only when the coverage of green technology meets $N > N^*$.

Therefore, for the government's green contract strategy, we have the following proposition.

Proposition 2. In order to promote emission reduction, the green technology T_T provided by the government should meet the following conditions:

- (1) the amount of green technology to each enterprise should be satisfied

$$0 < T_T < \frac{N - \psi}{2},$$

- (2) the coverage of green technology should be met

$$N > N^* = \frac{(r + \delta)(4a\delta x - \psi)}{r + \delta - \beta\gamma}.$$

4.2. The case of green funding

In this subsection, only some enterprises could receive green funding. Assume that enterprises indexed $i = 1, 2, 3, \dots, M$, which could receive green funding, are green enterprises, and that the rest of the enterprises, indexed $i = M + 1, M + 2, M + 3, \dots, N$, are generic enterprises. The output of green enterprises is q_c , and that of general enterprises is q_0 .

In this way, the income of the two types of enterprises can be expressed as:

$$\pi_i = pq_c - cq_c - \alpha q_c + T_C q_c \quad i = 1, 2, \dots, M, \quad (13)$$

for green enterprises.

$$\pi_j = pq_0 - cq_0 \quad j = M + 1, M + 2, \dots, N, \quad (14)$$

for generic enterprises.

For the current total output $Q = Mq_c + (N - M)q_0$, the market price can be expressed as

$$p = p_0 - a(Mq_c + (N - M)q_0).$$

The optimal output q_c^* , q_0^* can be obtained from the first order conditions of (13) and (14) respectively.

$$\begin{cases} \frac{\partial \pi_i}{\partial q_c} = p_0 - c - \alpha + T_c - a(N - M)q_0 - 2aMq_c = 0, \\ \frac{\partial \pi_j}{\partial q_0} = p_0 - aMq_c - c - 2a(N - M)q_0 = 0, \end{cases}$$

which implies

$$\begin{cases} q_c^* = \frac{p_0 - c + 2(T_c - \alpha)}{3aM}, \\ q_0^* = \frac{p_0 - c - (T_c - \alpha)}{3a(N - M)}. \end{cases} \quad (15)$$

According to equation (15), it can be seen that not only the optimal output q_c of green enterprises is affected by the additional cost α of emission reduction and green funding T_c but also that the optimal output of general enterprises is affected by them. To be specific, green funding T_c has a positive effect on the output of green enterprises but a negative effect on the output of general enterprises. The impact of emission reduction costs α on the two types of enterprises is just the opposite.

Then, the pollution evolution equation becomes

$$\dot{x} = M\gamma q_c^* + (N - M)q_0^* - \delta x.$$

That is, the optimization goal of government is

$$\begin{aligned} \max_{T_c} \quad & \int_0^t [Q^* - MT_c q_c^* - \beta x] dt \\ \text{s.t.} \quad & \\ \dot{x} = & \frac{(\gamma + 1)(p_0 - c) + (2\gamma - 1)(T_c - \alpha) - 3a\delta x}{3a}, \end{aligned} \quad (16)$$

Similarly, the HJB function V_C of optimal problem (16) could be written as

$$\begin{aligned} rV_C(x) = \max_{T_c} \{ & [Mq_c^* + (N - M)q_0^* - MT_c q_c^* - \beta x] \\ & + \frac{\partial V}{\partial x} [M\gamma q_c^* + (N - M)q_0^* - \delta x] \}. \end{aligned} \quad (17)$$

Combining equation (17), the optimal green funding of government satisfies

$$\frac{\partial V(x)}{\partial T_c} = \frac{1}{3a} - \frac{\psi_c + 4T_c}{3a} + \frac{\lambda}{3a}(2\gamma - 1) = 0, \quad (18)$$

which means

$$\lambda = \frac{\psi + 4T_c - 1}{2\gamma - 1},$$

where $\psi = p_0 - C - 2\alpha^C$.

Take the partial derivative of x on both sides of formula (7) and $T_c = k_c x + b_c$ to obtain

$$\begin{cases} k_c = \frac{1}{2} + \frac{3a(r + 2\delta)}{2(2\gamma - 1)}, \\ b_c = \frac{-3a[2\beta(1 - 2\gamma)^2 + 8\delta(\gamma\phi + \theta) - \gamma\Delta] + (2\gamma - 1)\Delta}{24a\delta(2\gamma - 1)}, \end{cases} \quad (19)$$

where $\Delta = \psi + 4\theta + 2\gamma(\psi + 1) - 1$.

Green funding in form of equation (19) is also a feedback Nash equilibrium solution. However, unlike the green technology case, the amount of green funding is independent of M and N , the number of companies involved in reducing emissions.

Finally, the net profit of green enterprises is

$$\pi_i = \frac{(p_0 - c + 2T_c - 2\alpha)^2}{9Ma},$$

the net profit of generic enterprises is

$$\pi_0 = \frac{(p_0 - c - T_c + \alpha)^2}{9(N - M)a}.$$

The following proposition gives the optimal production plan of both two types of enterprises.

Proposition 3. In the green funding case, the optimal output q^* of each enterprise is

$$\begin{cases} q_c^* = \frac{p_0 - C + 2(T_c - \alpha^C)}{3aM}, \\ q_0^* = \frac{p_0 - C - (T_c - \alpha^C)}{3a(N - M)}. \end{cases}$$

Based on q_i^* , the net profit π of each enterprise is

$$\begin{cases} \pi_c^* = \frac{(p_0 - c + 2T_c - 2\alpha)^2}{9Ma}, \\ \pi_0^* = \frac{(p_0 - c - T_c + \alpha)^2}{9(N - M)a}. \end{cases}$$

The emission of each enterprise E^* is

$$\begin{cases} E_c^* = \frac{(p_0 - c + 2T_c - 2\alpha)^2}{9Ma}\gamma, \\ E_0^* = \frac{(p_0 - c - T_c + \alpha)^2}{9(N - M)a}. \end{cases}$$

The total emission TE^* in the game is

$$\begin{cases} TE_c^* = \frac{(p_0 - c + 2T_c - 2\alpha)^2}{9a}\gamma, \\ TE_0^* = \frac{(p_0 - c - T_c + \alpha)^2}{9a}. \end{cases}$$

For shadow price λ , it from $\lambda < 0$ that

$$\lambda = \frac{\psi + 4T_c - 1}{2\gamma - 1} < 0,$$

the solution is

$$0 < T_c < \frac{1 - \psi}{4}. \quad (20)$$

By comparing equation (12), it can be seen that in order to ensure the promotion effect of green contracts on emission reduction, the amount of green funding is much smaller than that of green technology.

In the same environment, the total optimal emission $M\gamma q_c^*$ of green enterprises should be lower than the total optimal emissions $(N - M)q_0^*$ of general enterprises, which could reflect the advantages of green enterprises. According to $M\gamma q_c^* < (N - M)q_0^*$, the approximate range of the optimal green funding can be obtained

$$T_c > \frac{1 - \gamma}{2\gamma + 1}(p_0 - c) + \alpha^C.$$

The following statement summarizes the characteristics of green funding provided by the government.

Proposition 4. When $\frac{p_0 - c}{1 - 2\alpha^C} < \frac{2\gamma + 1}{5 - 2\gamma}$, there exists an optimal amount of green funding that satisfies the promotion of emission reduction and the advantages of green enterprises:

$$\frac{1 - \gamma}{2\gamma + 1}(p_0 - c) + \alpha^C < T_c < \frac{1 - \psi}{4}.$$

When $\frac{p_0 - c}{1 - 2\alpha^C} > \frac{2\gamma + 1}{5 - 2\gamma}$, we have

$$T_c < \frac{1 - \psi}{4} < \frac{1 - \gamma}{2\gamma + 1}(p_0 - c) + \alpha^C.$$

At this point, green funding can only promote the development of emission reduction, while the total emissions of green enterprises are higher than the total emissions of ordinary enterprises.

5. Green funding versus green technology

In this section, we will compare green funding and green technology

in terms of optimal emissions, pollution control and corporate returns, etc. The applicable scope and mutual transformation conditions of green funding and green technology are also given.

In order to better compare the advantages and disadvantages of green funding and green technology, the emission factor γ in the two cases is assumed to be the same. In addition, the green funding or green technology received by green enterprises is equivalent, i.e., $T_C = T_T = T$.

Let $E_T^* - E_c^* < 0$, we have

$$0 < \frac{M}{N} < \frac{2[p_0 - c + 2(T - \alpha)]}{3(p_0 - T - c - \alpha)},$$

and $TE_T^* - (TE_c^* + TE_0^*) < 0$, we obtain

$$T - \alpha < p_0 - c.$$

Proposition 5. Under the condition $T_C = T_T = T$, we have
For the optimal emissions of each green enterprises, we have

(1) when the coverage of green funds meets

$$0 < \frac{M}{N} < \frac{2[p_0 - c + 2(T - \alpha)]}{3(p_0 - T - c - \alpha)},$$

the emission of each green enterprise with green funding is higher than that with green technology, i.e.,

$$E_T^* < E_c^*,$$

(2) when the coverage of green funds meets

$$\frac{2[p_0 - c + 2(T - \alpha)]}{3(p_0 - T - c - \alpha)} < \frac{M}{N} < 1,$$

the emission of each green enterprise with green funding is lower than that with green technology, i.e.,

$$E_T^* > E_c^*,$$

For the total emissions in the game.

(3) when $T - \alpha < p_0 - c$, the total emissions with green funding is higher than that with green technology, i.e.,

$$TE_T^* < TE_c^* + TE_0^*,$$

(4) when $T - \alpha > p_0 - c$, the total emissions with green funding is lower than that with green technology, i.e.,

$$TE_T^* > TE_c^* + TE_0^*.$$

The proposition shows the following results:

Results (1) and (2) show that, for each green enterprise, green funding is more effective than green technology in terms of reducing emissions when the coverage of green funds reaches a certain value, i.e., $\frac{2[p_0 - c + 2(T - \alpha)]}{3(p_0 - T - c - \alpha)} < \frac{M}{N} < 1$; otherwise, green technology is better than green funding.

Results (3) and (4) show that, in terms of overall emissions control, total emissions in the game are not related to the coverage of green assistance but only to the amount of green assistance. Specifically, when the unit benefit of emissions reduction is smaller than the unit benefit of commodity sales, i.e., $T - \alpha < p_0 - c$, the total emission reduction caused by green technology is better than that caused by green funding.

As a conclusion, when the government aims to promote emission reduction, green technology is more effective than green funds, if and only if the coverage of green funds satisfies $0 < \frac{M}{N} < \frac{2[p_0 - c + 2(T - \alpha)]}{3(p_0 - T - c - \alpha)}$, and its

size satisfies $T - \alpha < p_0 - c$. Otherwise, green funding is more effective than green technology.

Then, we will compare the net profit of enterprises with green technology and green funding to find the critical conditions for players adopting green technology (or green funding).

Let $\pi_T - \pi_C < 0$, we have

$$\frac{(p_0 + T - c - \alpha)^2}{4Na} < \frac{(p_0 - c + 2T - 2\alpha)^2}{9Ma},$$

which means

$$0 < \frac{M}{N} < \frac{4[p_0 - c + 2(T - \alpha)]^2}{9(p_0 - T - c - \alpha)^2}.$$

Proposition 6. Green enterprises will choose green funding rather than green technology in a mitigation game if and only if the coverage of green funds meets M satisfies

$$0 < \frac{M}{N} < \frac{4[p_0 - c + 2(T - \alpha)]^2}{9(p_0 - T - c - \alpha)^2}.$$

Proposition 6 shows that, when the coverage ratio of green funding is within a certain range, i.e., $0 < \frac{M}{N} < \frac{4[p_0 - c + 2(T - \alpha)]^2}{9(p_0 - T - c - \alpha)^2}$, the benefits of green enterprises with green funding are greater than those with green technologies. That is, in this case, green enterprises are more inclined to receive green funding.

Combining **Proposition 5** and **Proposition 6**, it can be seen that there is a certain contradiction between the profit target of green enterprises and the development target of emission reduction. When the coverage of green funds is small, the benefits of green enterprises are larger, but the emission reduction effect is smaller than that of green technologies.

Therefore, the government needs to find the right coverage of green funding to ensure that corporate profit targets and emission reduction targets are met simultaneously. Let

$$\frac{4[p_0 - c + 2(T - \alpha)]^2}{9(p_0 - T - c - \alpha)^2} > \frac{2[p_0 - c + 2(T - \alpha)]}{3(p_0 - T - c - \alpha)},$$

we have

$$T + c - p_0 - \alpha > 0.$$

Proposition 7. When $T - \alpha > p_0 - c$, there exists optimal coverage of green funding to ensure that corporate profit targets and emission reduction targets are met simultaneously:

$$\frac{2[p_0 - c + 2(T - \alpha)]}{3(p_0 - T - c - \alpha)} < \frac{M}{N} < \frac{4[p_0 - c + 2(T - \alpha)]^2}{9(p_0 - T - c - \alpha)^2}.$$

Under the condition of **proposition 7**, both the government and green enterprises will be more inclined to choose green funding; otherwise, they will prefer green technology.

6. Example

In this section, we will provide numerical examples to demonstrate the effectiveness of the proposed results.

Consider an economy with one local government and $N = 10$ homogeneous enterprises, among which there are M green enterprises. The market price of the product p is $p = p_0 - a(Mq_c + (N - M)q_0)$, where $p_0 = 3$, $a = 1.2$, then $p = 3 - 1.2(Mq_c + (10 - M)q_0)$. Other parameters are given in **Table 1**.

The income of green enterprises is

Table 1
The basic parameters.

r	c	α	β	δ	γ	p_0	a
0.5	1	0.7	0.2	0.15	0.9	3	1.2

$$\pi_c = 1.8q_c - 1.2Mq_c^2 + 1.2(10-M)q_0q_c.$$

The income of generic enterprises is

$$\pi_g = 2q_0 - 1.2Mq_cq_0 + 1.2(10-M)q_0^2.$$

The pollution evolution equation becomes

$$\dot{x} = 0.9Mq_c^* + (10-M)q_0^* - 0.1x.$$

First, we show the pollution trajectories under an optimal production plan in two cases.

Fig. 1 shows the evolution characteristics of greenhouse gas accumulation in the atmosphere under green funding and green technology. It can be seen that at the initial stage of the game, the decline rate of greenhouse gases is roughly the same, which indicates that the efficiency of green funding and green technology in pollution control is the same at this stage. As the game progresses, the efficiency of green technology is gradually better than that of green capital. Eventually, both of the trajectories stabilize at a low level. The difference is that the stable-level for green technology is smaller than for green funding. This suggests that green technologies are more efficient in reducing emissions in the long run.

Then, the optimal green contracts for the two games is shown in Fig. 2.

Fig. 2 gives the optimal green contracts of the government for the two cases respectively. The optimal value of green technology is lower than green funding from the beginning to the end.

Combined with Figs. 1 and 2, it can be seen that, in the case of green technology, the financial pressure borne by the government is smaller. For greenhouse gas control, however, green technology is better than green funding. Therefore, we can conclude that green technology is superior to green funding in terms of unit emission reduction efficiency.

Then, we give a THREE-DIMENSIONAL image of the emission accumulation x with respect to time t and contracts T .

It can be seen that the cumulative amount of emissions decreases with time. However, for contracts T , there is not a simple decreasing relationship between emission accumulation x and contracts T . As shown in Fig. 3, the cumulative amount of emissions x at $T = 10$ is far greater than that at $T = 0$, which is consistent with the results of Propositions 2 and Proposition 4 (see Fig. 4).

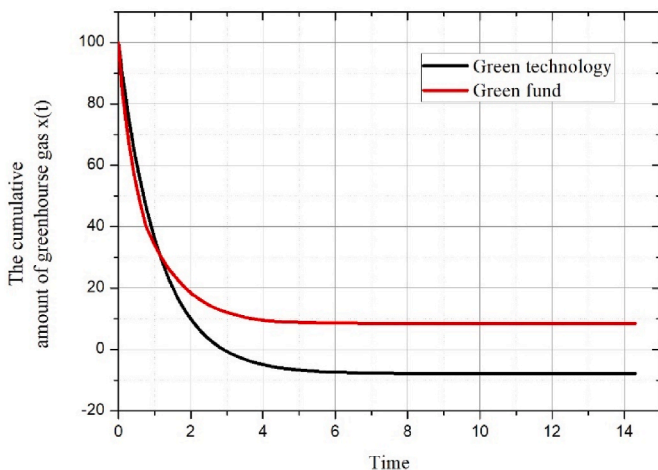


Fig. 1. The pollution trajectories of two cases.

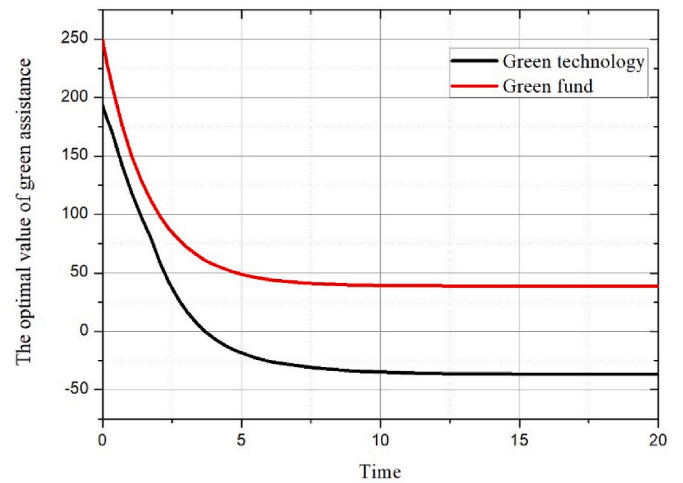


Fig. 2. The optimal green contracts of the government for the two cases. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

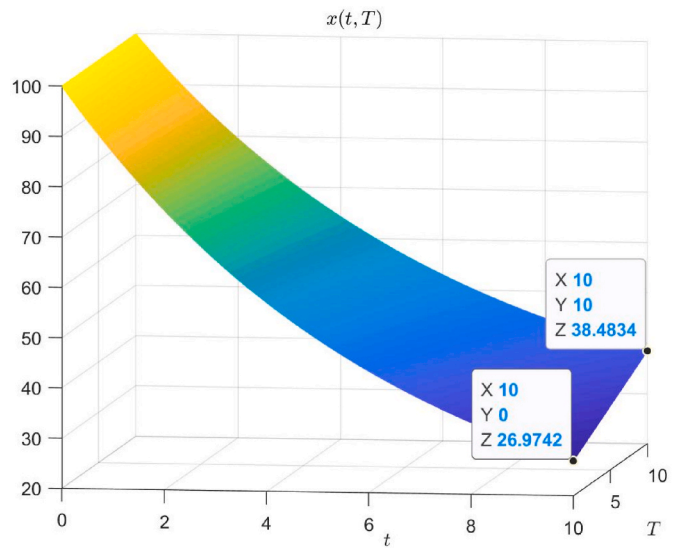


Fig. 3. The THREE-DIMENSIONAL image of the emission accumulation.

Next, we will discuss the advantages and disadvantages of green technology and green funding when $T_T = T_C$. It should be noted that $M = 10$ represents the case of green technology, while for green funding, we set four scenarios, i.e., $M = 2, M = 4, M = 6$ and $M = 8$.

According to proposition 5, the optimal emissions of green enterprises in five cases are given in Table 2.

Table 2 shows that the optimal emission of green enterprises has two features: (1) the instantaneous emission of green enterprises increases with the T , which means green contract has a positive impact on green enterprises' mitigation strategies; (2) when $T_T = T_C < 0.4$, there exists $M \in [0, N]$, i.e., $M = 6, 8$ so that the optimal emission of green enterprises with green funding is lower than those with green technology, according to proposition 5. When $T_T = T_C = 0.5$, we have $\frac{2[p_0 - c + 2(T - a)]}{3(p_0 - T - c - a)} > 1$, which no longer satisfies the condition in proposition 5, so whatever the value of M is, the optimal emissions of green enterprises with green funding is always larger than those with green technology.

Then, the relationship between the total optimal emissions of green enterprises and green contracts T is given in Fig. 3.

Fig. 3 shows that, with the increase of the amount of green contracts T , the output of green enterprises in both cases is monotonically

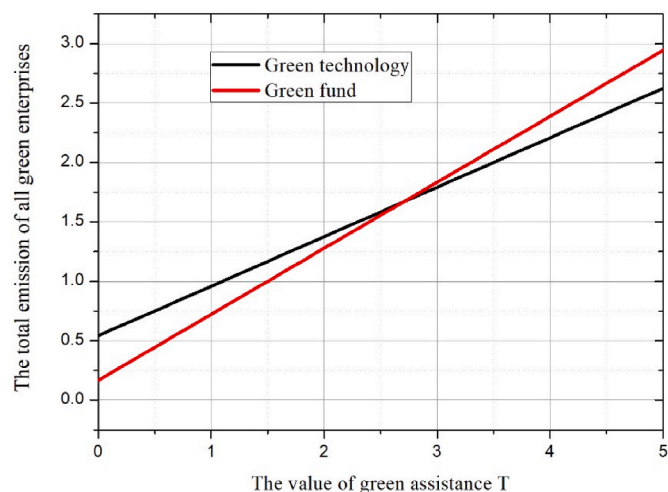


Fig. 4. The total optimal emissions of green enterprises. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2

The optimal emissions of green enterprise.

T	$M = 2$	$M = 4$	$M = 6$	$M = 8$	$M = 10$
E^*					
M					
0	0.083	0.041	0.027	0.021	0.054
0.1	0.111	0.055	0.037	0.027	0.058
0.2	0.138	0.069	0.046	0.034	0.062
0.3	0.166	0.083	0.055	0.041	0.066
0.4	0.194	0.097	0.064	0.048	0.070
0.5	0.222	0.111	0.074	0.055	0.075

increasing. This shows that green contracts can effectively motivate green enterprises to produce and reduce emissions. When the amount of green contracts $T = 2.7$, the total emissions of green enterprises in the two situations are the same, that is, their emission reduction efficiency is the same. When $0 < T < 2.7$, the emission reduction efficiency brought by green funding is better than that of green technology; When $T > 2.7$, the opposite is true. This is consistent with the conclusion of [proposition 5](#).

At last, the instantaneous profits of green enterprise in five cases are given in [Table 3](#).

[Table 3](#) shows that the instantaneous profits of green enterprises has two features: (1) the instantaneous profits of green enterprises increase with T ; (2) when $T_T = T_C$, there exists $M \in [0, N]$, i.e., $M = 8$ so that the instantaneous profits of green enterprises with green funding is higher than those with green technology, according to [proposition 6](#).

Combined with [Table 2](#), we can draw a conclusion: under the numerical example considered in this paper, when $0 < T < 0.5$ and green funding coverage meets $M = 8$, the reduction efficiency of green enterprises with green funding is higher than those with green technology. At the same time, green enterprises with green funding are more profitable. That is, both the government and green enterprises tend to choose green funding.

7. Conclusion

This paper study a differential transboundary pollution problem consisting of one local government and N homogeneous enterprises. In addition, the government provides two type of green assistance, i.e., green funding and green technology, to encourage enterprises to reduce their emissions. According to the different characteristics of the two

Table 3

The instantaneous profits of green enterprise.

T	$M = 2$	$M = 4$	$M = 6$	$M = 8$	$M = 10$
E^*					
M					
0	0.004	0.005	0.008	0.036	0.035
0.1	0.007	0.009	0.014	0.049	0.041
0.2	0.011	0.015	0.023	0.056	0.046
0.3	0.016	0.022	0.033	0.066	0.053
0.4	0.022	0.030	0.045	0.090	0.060
0.5	0.029	0.039	0.059	0.118	0.067

kinds of green assistance, two Stackelberg differential game models are established. By analyzing the models, we find that both types of green assistance have positive effects on enterprises' emission reduction. When the amount of funding does not exceed the threshold, there exists an optimal coverage rate of green funding; the emission reduction efficiency and profit of green enterprises with green funding are higher than those with green technology, which makes both the government and enterprises prefer green funding.

When comparing the differences between green funding and green technology, this paper only considers their different coverage areas, and other assumptions are the same. In fact, the mechanism of action of the two is not the same. Green technology is an important way to acquire technology and provides the possibility for the secondary innovation of low-carbon technology. However, green technology will result in high initial investment costs and learning costs. Green funding can effectively make up for the high cost of green innovation, but it has a lag. Therefore, whether providing green funding and green technology to enterprises at the same time can further improve their emission reduction efficiency is a very interesting question, which will also be the focus of our next work.

CRediT authorship contribution statement

Yufeng Chen: Conceptualization, Paper discussion and improvement. **Liming Li:** Conceptualization, Methodology, Writing – original draft, All authors have read and agreed to the published version of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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